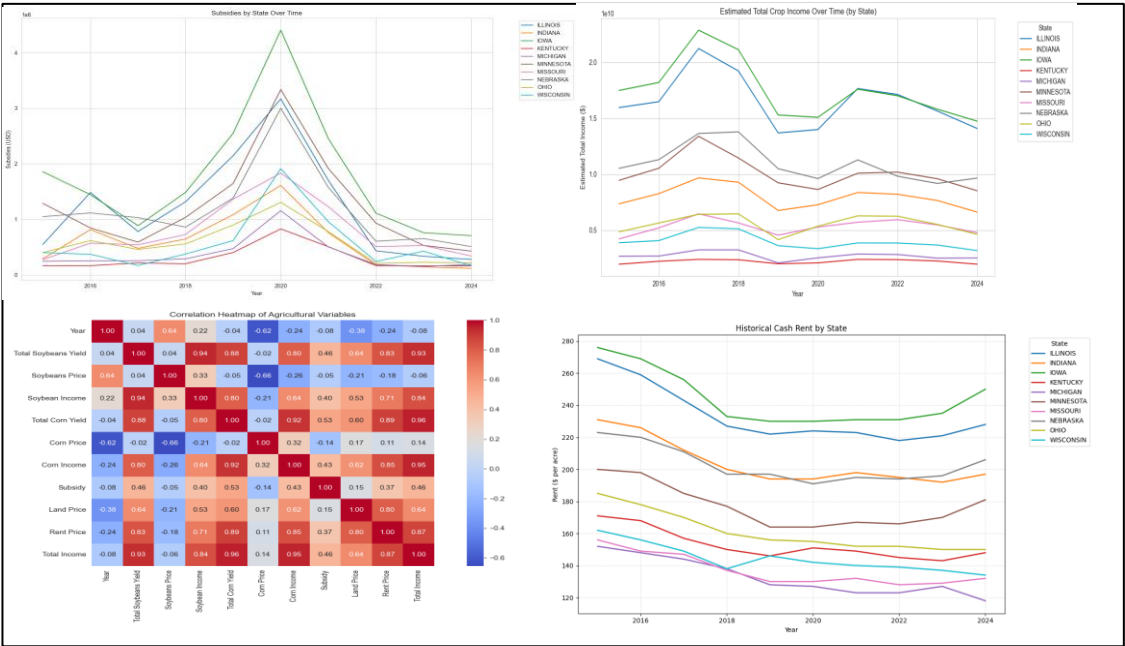


Following the Funds: The Relationship Between Farm Subsidies and State Economic Indicators in Row crop Country

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Farm subsidies make up a major portion of agricultural income in row-crop states, yet the distribution and economic impact of these payments are difficult to evaluate because the data are scattered, aggregated, and often non-specific to individual commodities. As a result, policymakers and the public have limited clarity on how subsidy levels relate to farm income trends across states. This project addresses that gap by analyzing ten years of state-level subsidy payments alongside yield-based income estimates for corn and soybeans, revealing how strongly (or weakly) subsidies align with actual agricultural performance



My dashboard was built by transforming a dense, single-sheet dataset into a visual system that exposes patterns the raw numbers kept hidden. The process began with cleaning and standardizing the yearly state-level observations so each variable could be compared consistently. From there, line charts were used to highlight trends over time, an intuitive way to show shifts in subsidies, yields, and market indicators without overwhelming the viewer. Bar charts served as direct comparisons between states or categories, making disparities immediately visible. A correlation heat map was included to reveal statistical relationships between variables, providing a fast, color-coded sense of which factors tend to move together. Scatter plots and geographic heat maps were not used because they offered little interpretive value for this dataset. The chosen visual techniques collectively turn the spreadsheet into a readable narrative about how agricultural economics behaves over time, turning the dashboard into a tool that guides the viewer toward meaningful relationships and long-term dynamics.

The data shows that while total crop income varies widely from year to year across major row-crop states, federal subsidies follow a steadier upward path, acting as a financial shock absorber that keeps farm revenue from collapsing during low-income periods. Some states receive disproportionately high support relative to their income—most notably Iowa—suggesting that federal programs reward historical acreage, risk profiles, and program structures rather than pure economic need. Cash rent, meanwhile, rises almost every year regardless of whether farmers earned more or less, indicating that land values respond more to long-term expectations and subsidized stability than to short-term market performance. The correlation heat map confirms these relationships: income tracks prices and yields, rent tracks long-run trends, and subsidies correlate with income only loosely. These findings matter because they reveal how dependent modern agriculture is on stabilizing payments that aren't fully captured in public datasets like QuickStats, especially since crop-insurance subsidies are excluded. Understanding these relationships helps explain why land remains expensive, why farms survive volatile markets, and why federal support continues to shape the economics of rural America.



All data was gathered from USDA quickstats and all come from survey data
Total income was calculated by multiplying Total acres planted multiplied by average yield multiplied by average price
Subsidies are aggregated data of all subsidies paid in that state for that year
Subsidies are underinflated due to lack of crop insurance subsidy data

POSTER GUIDELINES

SIZE: 36" x 48"

WHO: Students are responsible for printing and bringing posters to the gala

WHAT: Make sure to include the BDAA logo and your personal information (name, major, project track)

DO NOT PRINT THIS SLIDE



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